

Cryptography - 5_12

Today we take a look at **Post-Quantum Crypto**.

The issue: Quantum TMs are bad for cryptography (at least in theory):

- Shor invented an algorithm for factoring and DL that runs in poly time on a quantum machine/circuit.
- It requires many qubits and billions of quantum gates assuming a 2048-bit modules.

While we are still far from implementing Shor's algorithm, people believe it's just a matter of time. The NIST is worried. Until 5 years ago almost the entire crypto was factoring or DL based.

Not only that: quantum attacks might be already happening. I.E. "store now, break later".

Last but not least: Developing new crypto deployed in the real world takes ≈ 10 years.

For these reasons, the NIST started the standardization process for post-quantum crypto back in 2017.

Remark: PQ crypto means that Alice and Bob are still classica. Eve has a quantum computer.

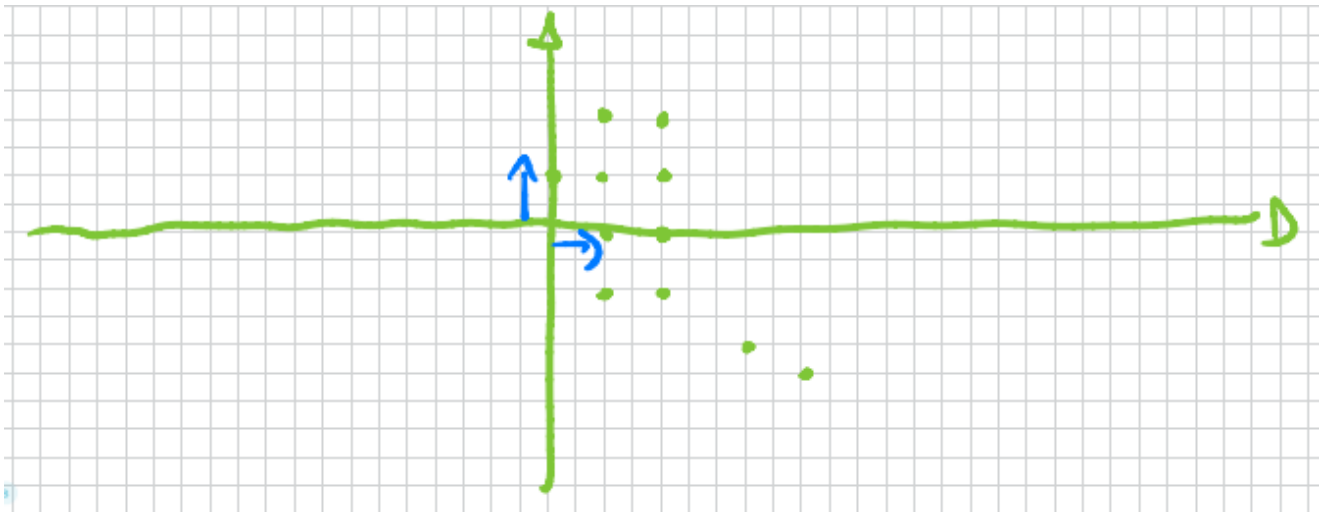
Minimal requirement: A computational problem believed to be hard for quantum machines. Many examples: Lattices, elliptic curves, cooles.

We'll focus only on **lattices**.

A lattice $\mathcal{L}$ is a discrete set of points in a n -dimensional space $\mathbb{R}^n$, generated by a linear combination of independent vectors.

Basis vectors $B = (\vec{b}_1, \dots, \vec{b}_n)$ over the reals.

$$\mathcal{L}(B) = B \cdot \mathbb{Z}^n = \left\{ \sum_{i=1}^n z_i \cdot \vec{b}_i : z_i \in \mathbb{Z} \right\}$$



Basis are not unique! For any $U \in \mathbb{Z}^{n \times n}$ unimodular (i.e. $\det(U) = \pm 1$), then $B \cdot U$ is also a basis because $U \cdot \mathbb{Z}^n = \mathbb{Z}^n$.

This is important as if $\det(U) = \pm 1$ we can map points of the lattice back and forth without leaving the integer set.

An important parameter for a lattice is the length of a shortest non-zero vector:

$$\lambda_1(\mathcal{L}) = \min_{\vec{v} \in \mathcal{L} \setminus \{\vec{0}\}} \|\vec{v}\|$$

where $\|\cdot\|$ is the euclidean norm, such that $\vec{v} = \sqrt{\sum v_i^2}$.

In general: $\lambda_i(\mathcal{L})$ is the i -th successive minima the smallest r such that $\mathcal{L}$ has i linearly independent vectors of length $\leq r$.

Here are some hard problems:

- SVP_γ : Given B , find a shortest non-zero vector $\vec{v} \in \mathcal{L}(B)$ such that $\|\vec{v}\| = \lambda_1(\mathcal{L})$ or $\leq \gamma(n) \cdot \lambda_1(\mathcal{L})$.
- GapSVP_γ : Given B just decide if $\lambda_1(\mathcal{L}) \leq 1$ or $\lambda_1(\mathcal{L}) > \gamma(n)$
- SIVP_γ : Given B , output $\{\vec{s}_i\}$ in linearly independent vectors such that $\|\vec{s}_i\| \leq \gamma(n) \cdot \lambda_n(\mathcal{L}) \forall i = 1 \dots n$

Fact: The only poly-time algorithm (even quantum) work for $\gamma(n) = 2^{O(n)}$.

The modern perspective: We will use equivalent assumptions:

✓ Definition (SIS)

Given m random vectors $a_i \in \mathbb{Z}_q^n$ forming matrix $A \in \mathbb{Z}_q^{m \times n}$, the $\text{SIS}_{n,q,\beta,m}$ is to find a non-zero integer $\vec{z} \in \mathbb{Z}^m$ of norm $\|\vec{z}\| \leq \beta$ such that

$$f_A(z) = A \cdot z = \sum_i \vec{a}_i \cdot z_i = \vec{0} \in \mathbb{Z}_q^n \pmod{q}$$

Remarks:

- SIS easy without the restriction on $\|z\|$. Also in case $\beta > q$, because $(q, 0 \dots 0)$ is a solution.
- Any solution w.r.t. A can be converted to solution for $[A|A']$ by appending zeroes.
- The values m, β must be large enough for a solution to exist. In particular, $\beta \geq \sqrt{\bar{m}}$ and $m \geq \bar{m} = \lceil n \log q \rceil$. First, wlog assume $m = \bar{m}$. Since there are more than $2^{\bar{m}} = q^n$ vectors $x \in \{0, 1\}^m$ and there must be two distinct x, x' such that $A \cdot \vec{x} = A \cdot \vec{x}' \in \mathbb{Z}_q^n$ and $\vec{z} = \vec{x} - \vec{x}' \in \{0, \pm 1\}^m$ is a solution of norm at most β .

$\implies f_A(\cdot)$ is automatically collision resistant.

Hardness: For $m = \text{poly}(n)$ and $q \geq \beta \cdot \text{poly}(n)$ solving $SIS_{n,q,\beta,m}$ is at least as hard as solving GapSVP_γ and SIVP_γ with $\gamma(n) = \beta \cdot \tilde{O}(\sqrt{n})$

✓ Definition (LWE)

For $\vec{s} \in \mathbb{Z}_q^n$, the LWE distribution $A_{\vec{s}, \mathcal{X}}$ over $\mathbb{Z}_q^m \times \mathbb{Z}_q$ is obtained by sampling $\vec{a} \in \mathbb{Z}_q^n$ random and $e \leftarrow \mathcal{X}$ and outputting:

$$\vec{a}, b = \langle \vec{s}, \vec{a} \rangle + e \pmod{q}$$

given m samples $(\vec{a}_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ from $A_{\vec{s}, \mathcal{X}}$ for random $\vec{s}$, find $\vec{s}$.

SEARCH – LWE _{$n,q,\mathcal{X},m$}

Remarks:

- Without noise, the problem is easy.
- Error distribution: $\mathcal{X}$ is taken to be any distribution such that $\Pr[|e| > \alpha \cdot q : e \leftarrow \mathcal{X}] \leq \text{negl}(\lambda)$ for $\alpha \ll 1$